THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Red blood cells swell and burst in distilled water. Reason (R): Water enters the cells by osmosis because they lack a cell wall. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A cell is placed in a solution and its mass increases by 20 per cent. The solution is: 1

(a) hypertonic **(b)** isotonic
(c) hypotonic to the cell **(d)** saturated salt

3. Which of the following could not be a function of the Golgi apparatus? 1

(a) synthesis of ATP **(b)** secretion
(c) packaging proteins **(d)** formation of lysosomes

4. The structure that gives the plant cell its definite shape is the: 1

(a) cell wall **(b)** vacuole **(c)** nucleus **(d)** cell membrane

5. Assertion (A): Plant cells do not burst when placed in distilled water. Reason (R): The cell wall resists excessive expansion of the cell. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Two cells A and B are placed in the same solution. A shrinks and B swells. This shows that the solution is: 1

(a) hypertonic to A and hypotonic to B **(b)** isotonic to both
(c) hypotonic to both **(d)** hypertonic to both

7. A student finds a cell with no nucleus, a cell wall and DNA free in the cytoplasm. The cell is: 1

(a) fungal **(b)** plant **(c)** bacterial **(d)** animal

8. In plant cells, the nucleus is pushed to the periphery by the: 1

(a) chloroplast **(b)** large central vacuole
(c) Golgi **(d)** ER

9. Assertion (A): Mitochondria are called semi-autonomous organelles. Reason (R): They contain their own DNA and ribosomes. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. A cell with 20 chromosomes divides by mitosis and its daughter cells again divide by meiosis. The number of chromosomes in the final cells is: 1

(a) 40 **(b)** 5 **(c)** 10 **(d)** 20

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do cells divide when they reach a certain size? Use the surface area to volume ratio for a cube of side 1 cm and 3 cm. 2

12. State why the cell membrane is described as a fluid mosaic. 2

13. How can you show that plant cells are plasmolysed in salt solution under a microscope? 2

14. A cell with 46 chromosomes undergoes mitosis twice. How many cells and how many chromosomes in each? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A cube of potato of mass 10 g is placed in 0.5 per cent, 1 per cent and 5 per cent sugar solutions. Predict the changes with the reasoning and mention the solution that is isotonic if the mass is unchanged in the 1 per cent solution. 3

16. A researcher wants to isolate pure mitochondria from liver cells. Describe the method in outline (centrifugation). 3

17. Describe how an artificial cell membrane (dialysis tubing) is used to demonstrate osmosis with glucose and starch. 3

OR

Explain the differences between the cell division in the root tip and in the testis. 3

18. Explain how the structure of the chloroplast allows it to trap light and make glucose. 3

OR

Explain the term totipotency and its relation to cloning plants by tissue culture. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an experiment to find the concentration of the cell sap of potato tissue: test cubes in 0, 2, 4, 6, 8 per cent sugar solutions, mention what to measure, expected results and the use of the graph (zero change in mass). 5

OR

Explain cell specialisation and the differentiation of stem cells with examples from plants and animals, and discuss medical uses and ethical issues. 5

20. Explain, with reference to the cell theory, the structure and function of the prokaryotic cell, eukaryotic cell and viruses, and the evolution of eukaryotes through endosymbiosis. 5

OR

A sample of cells is found to contain no nucleus, no mitochondria but has haemoglobin. Identify the cell, state its adaptations and the disadvantages of lacking those organelles. 5

END OF QUESTION PAPER — Total: 40 Marks